

Effects of chitosan oligosaccharide (COS) on the glycerol-induced acute renal failure *in vitro* and *in vivo*

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Received 19 December 2006; accepted 19 September 2007

Abstract

The purpose of this study was to investigate the effects of chitosanoligosaccharide (COS) on the change of inflammatory response, renal function factor, and renal oxidative stress in glycerol induced ARF *in vitro* and *in vivo*.

The molecular weight of COS was approximately below 10 kDa with 90% degree of deacetylation. Renal proximal tubular cells were treated with only COS (0, 0.01, 0.025, 0.05, 0.075 and 0.1%) or COS in the presence of glycerol (4 mM). And rats were administered with glycerol (50%, 8 ml/kg) by intramuscular injection for the induction of ARF. For identification the protection effect of COS in the glycerol induced ARF, rats were administered by COS (0.05 and 0.1%) using P.O. injection. The enzymatic activity of the released RDPase was assayed by the fluorometric method. The level of TNF α in kidney or the culture medium was quantified using ELISA kit (R&D Systems, Minneapolis, USA) and, nitrite concentration was determined by the Griess reaction.

We showed that COS stimulated the production of TNF α , NO and the released RDPase. Glycerol increased the concentration of RDPase in kidney and decreased the released RDPase in proximal tubular cells. And, glycerol increased the production of NO, TNF α , creatinine, and MDA, and decreased SOD. However, COS recovered the glycerol induced inflammatory response, renal function factor, and antioxidant effect in kidney.

COS had the antioxidant activity and the anti inflammatory effect. And maybe that characteristics could help recover the glycerol induced ARF.

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Keywords: Chitosanoligosaccharide (COS); Glycerol; Nitric oxide (NO); TNF α ; Renal dipeptidase (RDPase); Superoxide dismutase (SOD); Malondialdehyde (MDA)

1. Introduction

Glomerular and interstitial macrophage infiltration is a feature of all types of acute injury and chronic progressive renal disease (Nikolic-Paterson et al., 1997; Heather et al., 2004). Acute renal failure (ARF) associated with ischemic

or hypoxic injury or exposure to nephrotoxic substances. ARF leads to reduction in salt and water excretion, reduction in the urea clearance, reduction in renal blood flow and glomerular filtration rate, induction elevation of plasma ciprofloxacin levels and change the renal apparent diffusion coefficient (ADC). There are three major goals in the management and treatment of the patient with ARF. The first is to minimize the extent of the parenchymal injury, the second is to prevent the symptoms of

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uremia, and the third is to promote more rapid and complete recovery (William, 1990; Naora et al., 1999; Vladimir et al., 1996). ARF is usually reversible, the kidney being able to recover from almost complete loss of function. Nevertheless, ARF is associated with major inpatient morbidity and mortality, reflecting the severity of the casual illnesses and the high frequency of complications (Brady and Singer, 1995).

Glycerol is a well known material for the induction of ARF *in vivo*. It is reported that the acute volume depletion models of glycerol-induced ARF are more closely related to the syndrome of ARF in humans than is the chronic dehydration model (Yehuda, 1993). Glycerol-induced myohemoglobinuric ARF shows many of the hallmarks of the 'crush syndrome' the archetypical form of human ARF that first led to widespread recognition of this disorder in man (Allen and Donald, 1989). Also, glycerol promotes free radical formation, lipid peroxidation, and renal dysfunction (Halliwell and Gutteridge, 1990).

Tumor necrosis factor- α (TNF- α) is a member of TNF ligand family that exerts their biological activities by interacting with four receptors of the TNF family. Macrophages, lymphocytes, neutrophils, and mast cells produce this inflammatory cytokines. TNF- α is known to be a key mediator for the induction of apoptosis and development of humoral immune response. TNF- α is a proinflammatory cytokines. However, at high concentrations, TNF- α has disadvantageous effects, such as inducing tissue injury and potentiating septic shock (Wallach et al., 1999; Shohami et al., 1999; Ettinger et al., 1998). TNF- α may have homeostatic effect in limiting the extent of an inflammatory response as well as acting as an antimalarial agent and function in intra-membranous bone repair (Gary and Shaomian, 2003; Marino et al., 1997; Ma, 2001; Gerstenfeld et al., 2001).

Nitric oxide (NO) is a free radical generated by NO synthases (NOS). NO has a short half-life and is rapidly oxidized to the stable, inactive end-products, nitrite and nitrate. NO may contribute to many pathologic disorders such as autoimmune disease, tumor growth, and metastasis, and septic shock (Moncada, 1997; Moncada et al., 1991).

Renal dipeptidase (RDPase, membrane dipeptidase, dehydropeptidase 1, EC 3.4.13.19) has been widely studied since it was firstly purified from porcine kidney brush border membrane (Campbell et al., 1966). RDPase is a zinc-metalloenzyme principally localized in the brush-border of the renal proximal tubule (Welch and Campbell, 1980). The dipeptidase has been proposed as a diagnostic marker enzyme because its activity in urine samples of acute renal failure (ARF) and chronic renal failure (CRF) patients decreases (Ito et al., 1984; Lee et al., 1999; Fukumura et al., 1999).

Chitosan, the linear polymer of D-glucosamine in β (1–4) linkage, has been recommended as a suitable functional material because of its biocompatibility, biodegradability, non-toxicity, adsorption properties and regulation cell activation (Kumar, 2000; Yoon et al., 2005; Pae et al., 2001).

In this study, we investigate the effects of COS on inflammatory, renal function as well as renal oxidative stress in the glycerol-induced ARF in rat.

2. Materials and methods

2.1. Materials

Ala Ala, Ala dehydrogenase, β NAD⁺, L Arg, diaphorase and resazurin were purchased from the Sigma Aldrich Chemical Company (St. Louis, MO, USA). Dulbecco's modified eagle's medium (DMEM), fetal bovine serum (FBS), 100 μ l/ml streptomycin and 100 units/ml penicillin were purchased from Gibco BRL (Grand Island, NY, USA), Tissue culture plate and 100 mm diameter dishes were purchased from Nunc (Naperville, IL). Bicinchoninic acid (BCA) protein assay kit was purchased from Pierce (USA). Sodium nitrate, sulfanilamide and *N* (naphthyl) ethylene diamine were purchased from Sigma (USA). Chitosan oligosaccharide (M.W. 10 kDa, deacetylation degree > 90%) was obtained from EcoBio (Gwangju, Korea). TNF α kits were purchased from R&D system (Minneapolis, MN, USA). SOD and MDA kits were purchased from CALBIOCHEM (San Diego, USA).

2.2. Cell culture

Proximal tubular cells were grown in DMEM with 10% (v/v) FBS, 100 μ l/ml streptomycin and 100 units/ml penicillin and incubated at 37 °C in humidified atmosphere containing 5% CO₂. Cells were grown to be 80% confluence in sterile tissue culture flasks and gently detached by scraping. Cell number counting using trypan blue exclusion, was done with a haemocytometer. For the *in vitro* study of COS, cells were cultured in triplicate at a density of 1×10^6 cells/ml in 12 well tissue culture plates for 24 at 37 °C in 5% CO₂ incubator. Then cells were treated with various concentrations of COS solution with or without glycerol (4 mM). Culture supernatants were removed for nitrite, TNF α measurement 6, 12 or 24 h after activation.

2.3. Animals and treatment

The rats were purchased from Samtako (Korea). Rats were housed in animal cage, kept at 25 °C, with 12 h light/dark cycle, and allowed free access to a standard chow diet and water *ad libitum*. Rats were divided into three groups. Group 1 was not treated with COS after intramuscular injection of glycerol (50%, 8 ml/kg) (Yehuda, 1993). Group 2 was treated with COS (0.05%) by P.O. after intramuscular injection of glycerol (50%, 8 ml/kg). And Group 3 was treated with COS (0.1%) by P.O. after intramuscular injection of glycerol (50%, 8 ml/kg). All groups were treated with COS for 12 h, 24 h, 48 h, 72 h, 96 h or 128 h ($n = 10$, each time). Each kidney was frozen in small aliquots at -80 °C after collection.

2.4. Nitrite assay

Nitric oxide synthesis was determined by assaying culture supernatants for nitrite, the stable reaction product of NO and molecular oxygen. Nitrite concentration was determined by the Griess reaction. Briefly, 100 μ l of culture supernatants were incubated with equal volumes of the Griess reagent (1% sulfanilamide/0.1% *N* (1 naphthyl) ethylenediamine dihydrochloride/2.5% H₃PO₄) at room temperature for 15 min and absorbance (540 nm) was determined in a spectrophotometer. Nitrite concentration was determined by using sodium nitrite (0.5 to 100 M) as a standard.

2.5. Measurement of TNF α

The production of TNF α secreted by proximal tubules or tissue samples was measured by ELISA kits. The supernatants or tissue samples were analyzed for TNF α by a using a commercial ELISA kit according to

the manufacturer's guidelines. And, 96 well plate were read at 450 nm using microplate reader (Molecular Devices, USA).

2.6. Spectrofluorometric assay of RDPase

The enzymatic activity of the released RDPase was assayed as described previously according to the fluorometric method (Ito et al., 1984). A typical 2.5 ml reaction mixture consisted of 1.5 μ M L-Ala-L-Ala, 0.4 U L-Ala dehydrogenase, 1.5 μ M β -NAD⁺, 0.06 U diaphorase and 0.1 μ M resazurin sodium in 12.5 mM Na₂CO₃ (pH 9.0) buffer. The incubation supernatant (20 μ l), containing the released RDPase was added last to the pre-incubated reaction mixture and the reaction was carried out for 2 min at 25 °C. The concentration of resorufin, the fluorescent product of the coupled reactions, was measured photometrically for 2 min using its excitation and emission maximums of 568 nm and 589 nm respectively using a JASCO spectrofluorimeter (JASCO Co., Hachioji City, Tokyo, Japan).

2.7. Preparation of porcine proximal tubules

The kidney was perfused with cold phosphate buffered saline (PBS) to completely remove the blood via a polyethylene catheter, which was inserted above the bifurcation of the renal artery. The renal outer sack was removed; the cortex was minced into small pieces and homogenized by one stroke each with a polytron tissue homogenizer and a teflon glass homogenizer (GlasCol, Terre Haute, IN, USA) set on low speed (25–30 rpm). The crude homogenate was passed through a 253 nm nylon mesh and then an 83 nm mesh (Tetko, Kansas City, MO, USA). The proximal tubules retained on the 83 nm mesh were washed with 10 volumes of cold PBS and then with an equal volume of Krebs Ringer Hepes (KRH) buffer consisting of 125 mM NaCl, 5 mM KCl, 1.2 mM KH₂PO₄, 1.2 mM MgSO₄, 2 mM CaCl₂, 6 mM glucose and 25 mM Hepes (pH 8.0). The scraped proximal tubules were then resuspended in 2 volumes of KRH buffer and centrifuged (600g, 5 min). The pellet was resuspended in 50 volumes of KRH buffer and was referred to as the 'proximal tubules'.

2.8. Protein assay

Samples (25 μ l) were transferred to a 96 well microplate and then 200 μ l of bicinchoninic acid (BCA) reagent was added. Bovine serum albumin was used as standard. After 1 h incubation at 37 °C, protein concentrations were determined from the absorbance at 540 nm using microplate reader (Molecular devices, USA).

2.9. Western blot analysis

Proteins (20 μ g) in the kidney lysates were then separated by 15% SDS PAGE and transferred to PVDF membrane. The membrane was blocked with 5% skim milk in TPBS for 1 h at room temperature and incubated with anti mouse TNF α antibody for 3 h at room temperature or overnight at 4 °C. After being washed in TPBS three times, the blot was incubated with secondary antibody (horseradish peroxidase conjugated anti goat antiserum) for 1 h at room temperature. The antibody specific proteins were detected by using West ZOL[®] (plus) (iNtRON Biotech nology Inc, Korea).

2.10. Creatinine assay

The creatinine concentrations were measured according to the method described in ASAN SET Creatinine kit (Asanpharm, Korea). Rats were treated with glycerol (50%, 8 ml/kg) by intramuscular injection and COS (0.1%) by P.O. for 48 h.

2.11. SOD assay

The method is based on the formation of red formazan from the reaction of 2 (4-iodophenyl) 3 (4-nitrophenol) 5-phenyltetrazolium chlo-

ride and superoxide radical (produced in the incubation medium from the xanthine xanthine oxidase reaction system), which is assayed spectrophotometrically at 505 nm. For the determination of SOD activity, 12.5 μ l of kidney tissue lysate was used. The inhibition of the produced chromogen is proportional to the activity of SOD present in the sample. A 50% inhibition is defined as one unit of SOD and the specific activity is represented as units per mg protein. Rats were treated with glycerol (50%, 8 ml/kg) by intramuscular injection and COS (0.1%) by P.O. for 48 h.

2.12. MDA assay

The malondialdehyde (MDA) content, a measure of lipid peroxidation, was assayed in the form of thiobarbituric acid reacting substances (TBARS). In brief, the reaction mixture consisted of 0.2 ml of 8.1% sodium lauryl sulphate, 1.5 ml of 20% acetic acid solution adjusted to pH 3.5 with sodium hydroxide and 1.5 ml of 0.8% aqueous solution of thio-barbituric acid was added to 0.2 ml of 10% (w/v) of PMS. The mixture was brought up to 4.0 ml with distilled water and heated at 95 °C for 60 min. After cooling with tap water, 1.0 ml distilled water and 5.0 ml of the mixture of *n*-butanol and pyridine (15:1 (v/v)) was added and centrifuged at 3000 rpm for 15 min. The organic layer was taken out and its absorbance was measured at 532 nm. TBARS were quantified using an extinction coefficient of 1.56×10^5 M⁻¹ cm⁻¹ and expressed as nanomoles of TBARS per milligram protein. The renal MDA content was expressed as nanomoles of MDA per milligram of protein. Rats were treated with glycerol (50%, 8 ml/kg) by intramuscular injection and COS (0.1%) by P.O. for 48 h.

2.13. Statistical analysis

All data were expressed as mean $\pm$ S.D. The data analysis was performed by ANOVA test. A *P* value of <0.05 was considered significant. The comparison was carried out with control or with reagent treated data.

3. Results

3.1. Effects of COS on RDPase release and activity in vitro and in vivo

In our previous study, we observed that effects of COS on released RDPase activity in proximal tubules. We showed that the RDPase release and activity was increased by the COS concentration dependent manner. Especially, in 0.05, 0.075 and 0.1% COS concentration, the released RDPase activity was significantly increased compared to other COS concentration (data not shown). We observed the recovery effect of COS on inhibition of RDPase activity by glycerol-induction. The released RDPase activity was decreased in proximal tubules by the glycerol concentration dependent manner (data not shown). However, the decreased RDPase activity was recovered by COS on glycerol-induced damage (Fig. 1. **p* < 0.05).

In Fig. 2, we observed that the effects of COS on RDPase activity in rat kidney. In the only-glycerol group, glycerol increased the RDPase in kidney. Whereas, in COS (0.1%) plus glycerol group, COS decreased the RDPase activity in kidney compared with only-glycerol group (**p* < 0.05). This result means that it was induced by COS that the release of RDPase was decreased by glycerol *in vivo*, too.

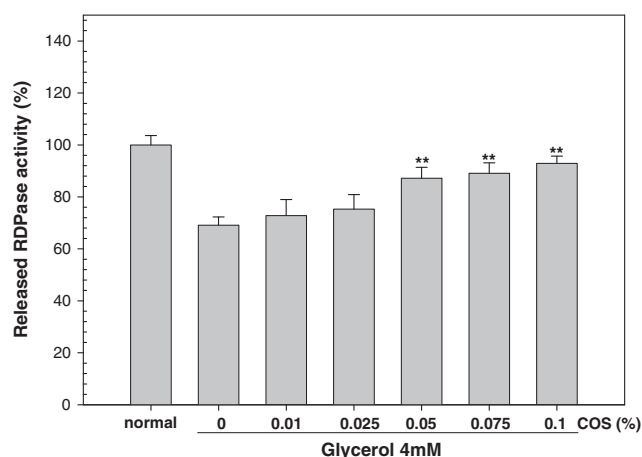


Fig. 1. Effects of chitosan oligosaccharide in the presence of glycerol on the release of RDPase from proximal tubules. Proximal tubular cells were incubated at 37 °C in KRH buffer alone (normal) and chitosan oligosaccharide (0, 0.01, 0.025, 0.05, 0.075, and 0.1%) with glycerol (4 mM). The RDPase released into supernatant was assayed as described on the methods. The results are expressed as percentage compared with the arbitrary intensity of glycerol (3–6 determinations, mean $\pm$ SD, * P < 0.05 vs. only glycerol).

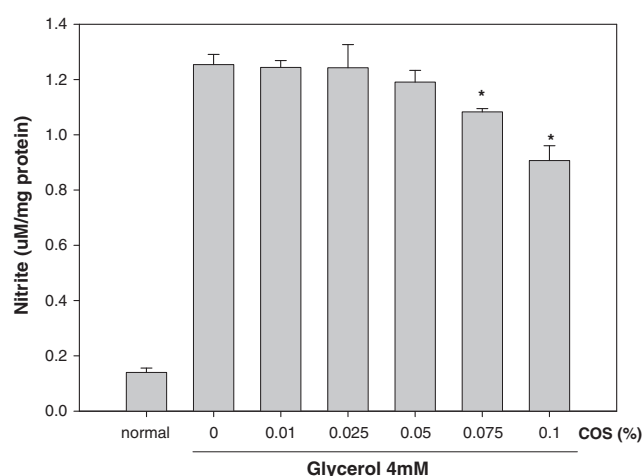


Fig. 3. Effects of chitosan oligosaccharide on the NO production in the presence of glycerol from proximal tubular cells. Proximal tubular cells were diluted 50 fold by KRH buffer and incubated for 10 min at 37 °C in the concentration dependent manner of chitosan oligosaccharide in the presence of glycerol. The Nitrite was assayed as described on the methods. The results are expressed as compared with the arbitrary intensity of normal (3 determinations, mean $\pm$ SD, * P < 0.05 vs. normal).

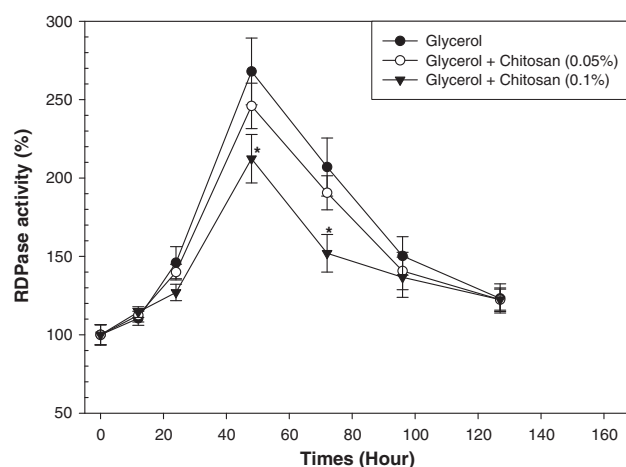


Fig. 2. Effects of chitosan oligosaccharide on the glycerol induced RDPase in the kidney. Glycerol (50%, 8 mL/kg) solution was injected by intramuscular injection in rats. 0.05, 0.1% of chitosan oligosaccharide was injected by intramuscular injection in rats. Each group was used to 10 rats (10 determinations, mean $\pm$ SD, * P < 0.05 vs. glycerol at each time).

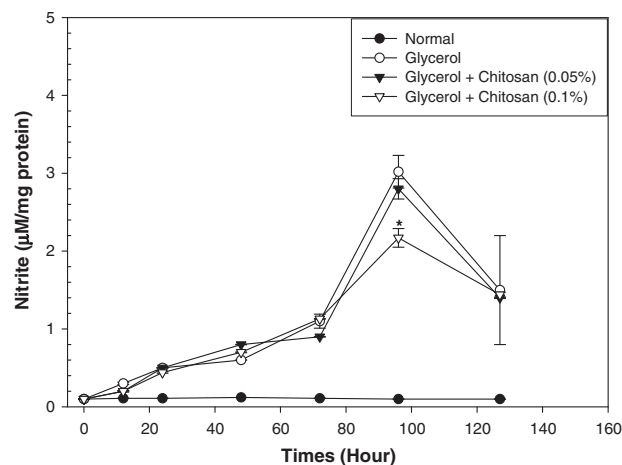


Fig. 4. Effects of chitosan oligosaccharide on the glycerol induced NO production in the kidney. Glycerol (50%, 8 mL/kg) solution was injected by intramuscular injection in rats. 0.05, 0.1% of chitosan oligosaccharide was injected by intramuscular injection in rats. Each group was used to 10 rats (10 determinations, mean $\pm$ SD, * P < 0.05 vs. glycerol at each time).

3.2. Effects of COS on NO production in vitro and in vivo

In our previous study, we observed that the effect of COS on NO production in proximal tubules. We showed that the NO production was significantly increased by the COS dependent concentration manner compared with normal (* p < 0.05) *in vitro* (Yoon et al., 2003). However, the glycerol induced NO production in proximal tubules was decreased by COS (Fig. 3. * p < 0.05).

In Fig. 4, we observed the effect of COS on NO production in rat kidney. There was no significant difference among all groups including control groups until 72 h.

But, at 96 h, in the only-glycerol group, glycerol increased the NO production. Whereas, in COS plus glycerol group, COS decreased the NO production compared with only-glycerol group (* p < 0.05).

3.3. Effects of COS on TNF- α production in vitro and in vivo

In Fig. 5, we observed that the effect of COS on TNF- α production in proximal tubules. We showed that the TNF- α production was increased by the COS concentration dependent manner by itself (* p < 0.05).

In Fig. 6, we observed the effect of COS on glycerol-induced TNF- α production in proximal tubules.

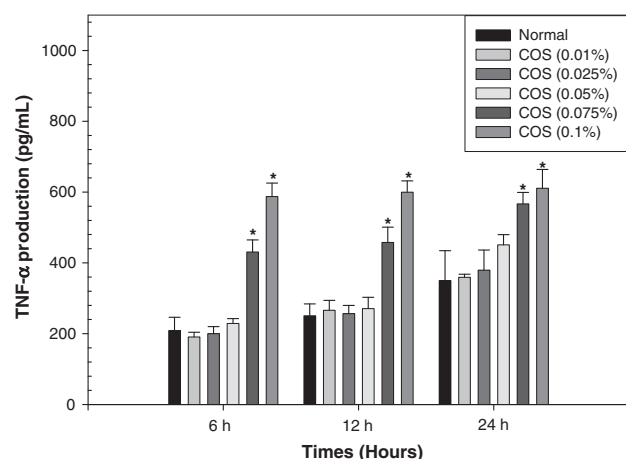


Fig. 5. Effects of chitosan oligosaccharide on the TNF α production from proximal tubular cells as a function of concentration. Each group was cultured during the 6, 12, or 24 h. And, TNF α released into the culture Medium was assayed by ELISA kit. The results are expressed as compared with normal (3 determinations, mean $\pm$ SD, * P < 0.05 vs. normal).

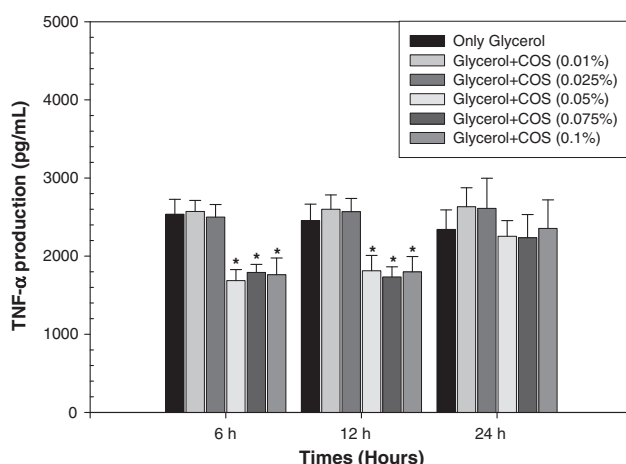


Fig. 6. Effects of chitosan oligosaccharide on the glycerol induced TNF α production in the proximal tubular cells. Proximal tubular cells were incubated at 37 °C in KRH buffer alone and chitosan oligosaccharide (0, 0.01, 0.025, 0.05, 0.075, and 0.1%) with glycerol (4 mM). The TNF α released into supernatant was assayed as described on the methods. The results are expressed as percentage compared with glycerol (3–6 determinations, mean $\pm$ SD, * P < 0.05 vs. only glycerol).

The production of TNF- α was highly increased by glycerol-administration about 10 folds compared with normal group. Whereas, in COS plus glycerol group, COS decreased the TNF- α production compared with only-glycerol group (* p < 0.05).

In Fig. 7, we observed the effect of COS on TNF- α production using ELISA and western blot analysis in rat kidney. Western blot was carried out using lysate sample at 48 h, because the effect of COS was significant. In the only-glycerol group, glycerol increased the TNF- α production compared with normal group. Whereas, in COS plus glycerol group, COS decreased the increased TNF- α pro-

duction by glycerol compared with only-glycerol group (* p < 0.05).

3.4. Effects of COS on glycerol-induced ARF using the serum creatinine concentration, and antioxidant alteration

Glycerol injection significantly increased the serum creatinine and MDA in kidney. COS significantly decreased serum creatinine and MDA in kidney (Table 1, * p < 0.05). Moreover, the SOD, which was markedly reduced by intramuscular injection of glycerol, was markedly improved by COS treatment (Table 1, ** p < 0.05). However, COS had no effect on serum creatinine, MDA, and SOD (Table 1).

4. Discussion

In this study, we evaluated the effects of COS on the renal function, inflammatory and antioxidant factors including the RDPase, serum creatinine, NO, TNF- α , MDA and SOD in the proximal tubules and rat kidney.

The induction of iNOS protein by cytokines such as TNF- α is known to relate to cell and/or tissue injury via the cytotoxic effect of the free radical, NO, produced in large quantities by various NOS, including NOS, iNOS and eNOS (Pinsky et al., 1995). It was reported that chitosan has an in vitro stimulatory effect on NO production (Peluso et al., 1994). The low molecular weight chitosan oligosaccharides (<1 kDa) could inhibit NO production in murine macrophage cell line, RAW 264.7, in the presence of strong inflammatory stimuli, bacterial lipopolysaccharide (Niranjan et al., 2006). In this study, we used 10 kDa COS and glycerol. NO production was increased by glycerol. However, COS decreased the glycerol-induced NO production in the proximal tubules and kidney tissue.

The increase and recovery of RDPase activity were suggested as an indication of strengthening of renal function and improvement of renal failure when the RDPase activity in urine sample of acute and chronic renal failure patients decreases (Fukumura et al., 1999). Glycerol is known to cause acute renal failure by severely damaging proximal tubule cells. We showed that the RDPase activity in kidney was increased by glycerol, and the released RDPase in proximal tubules was decreased by glycerol. However, COS was recovered the change of RDPase by glycerol-treatment in the proximal tubules and kidney tissue.

Pro-inflammatory cytokines are produced in response to a variety of stimuli including bacterial endotoxin, pro-oxidant, and certain chemical agents. Especially, Pro-inflammatory cytokines, including TNF- α , IL-1 and IL-6 participate in both physiological and pathological conditions. Increasing evidence reported that secretion of these products at low levels following tissue damage participate in cellular repair mechanism while, when produced in excess, pathological sequelae may ensue (Bertrand et al., 2004). Cytokine-mediated inflammatory processes have been implicated in the pathogenesis of these states.

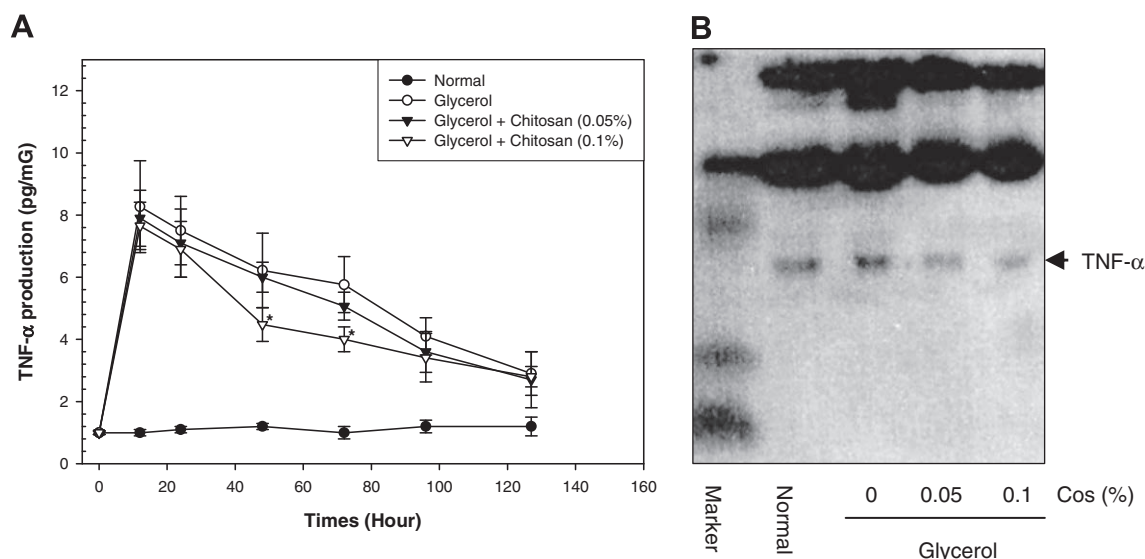


Fig. 7. Effects of chitosan oligosaccharide on the glycerol induced TNF- α production in the rat kidney. TNF- α in kidney was detected by ELISA method (A) in the time dependent manner and western blot analysis (B) at 48 h. Glycerol (50%, 8 mL/kg) solution was injected by intramuscular injection in rats. 0.05, 0.1% of chitosan oligosaccharide was injected by intramuscular injection in rats. Each group was used to 10 rats (10 determinations, mean $\pm$ SD, * P < 0.05 vs. glycerol at each time).

Table 1

Effects of chitosan oligosaccharide (COS) on the glycerol induced serum creatinine, superoxide dismutase (SOD) and malondialdehyde (MDA) in the rats

	Serum creatinine	SOD (U/mg protein)	MDA (nmol/mg protein)
Normal	0.6 $\pm$ 11	2.25 $\pm$ 0.39	33.0 $\pm$ 3.17
Glycerol	2.61 $\pm$ 1.30*	0.91 $\pm$ 0.12*	76.3 $\pm$ 5.35*
COS + Glycerol	1.42 $\pm$ 0.29**	1.95 $\pm$ 0.32**	47.5 $\pm$ 1.49**
COS	0.62 $\pm$ 0.18	2.27 $\pm$ 0.15	29.8 $\pm$ 4.16

Glycerol (50%, 8 mL/kg) solution was injected by intramuscular injection in rats. Chitosan oligosaccharide (0.1%) was injected by per oral in rats. Each group was used to 10 rats (10 determinations, mean $\pm$ SD, * P < 0.05 vs. normal, ** P < 0.05 vs. glycerol).

TNF- α initiates the inflammatory cascade, and induces the production of numerous additional mediators (Baud et al., 1994). TNF- α released from either infiltrating or intrinsic glomerular cells is thought to contribute to glomerular injury in experimental models of toxic or immunologic-mediated renal diseases as well as ischemia (Ziesche et al., 1994). The water soluble chitosan (<30 kDa, WSC) acts as an accelerator of RAW 264.7 macrophages activation by rIFN- γ via process involving L-arginine-dependent NO production. Then, RAW 264.7 macrophages were stimulated with rIFN- γ plus WSC, the expression of TNF- α protein (Jeong et al., 2000). The low molecular weight chitosan oligosaccharides (<3.5 kDa) decreased the production of proinflammatory cytokine (IL-1 α and TNF- α) (Kim et al., 2006). This study showed that the production of TNF- α was increased by glycerol in vitro and in vivo. However, COS recovered the production of TNF- α by glycerol in the proximal tubules and kidney tissue. TNF- α and IL-6 was significantly inhibited by pretreatment with

water-soluble chitosan (WSC) in human astrocytoma cells. And the expression of iNOS was partially inhibited by treatment by WSC (Kim et al., 2002). In this study, we demonstrate that TNF- α production in both proximal tubules and kidney tissue were increased by glycerol administration. The treatment of COS significantly inhibited the production of TNF- α in proximal tubules and rat kidney.

It was reported that glycerol caused the renal oxidative stress. Also, catechin had the protective effect on glycerol-induced ARF via antioxidant effect (Halliwell and Gutteridge, 1990; Vikas et al., 2003). This study showed that COS had similar antioxidant effect and renal function compared with catechin. These results showed that COS could reduce the glycerol-induced ARF via antioxidant effect.

Thus, this study shows clearly evidence of the potential protective effects of COS on the glycerol-induced ARF *in vitro* and *in vivo*.

Acknowledgements

This work was supported by Grant R01-2004-000-10536-0 from the Basic Research program of the Korea Science & Engineering Foundation.

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